Robotic Manipulation

Exercise 3 - Report

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- **Equations to compute the poses**

  PRE GRASP POSE = ${}^{base}T * {}^{base}T_{pick} * T_z(0.1) * R_x(\pi)$
  GRASP POSE = ${}^{base}T * {}^{base}T_{pick} * R_x(\pi)$
  PICK POSE = ${}^{base}T * {}^{base}T_{place} * R_x(\pi)$

- **In which coordinate frame the MoveIt assumes the poses are specified and what should I do if my pose is specified with respect to another frame?**

  The poses are specified, by default, with respect to the reference frame of the robot, the base frame. If you have a pose with respect to another frame, you have to compute the pose with respect to base frame with the proper matrix multiplications.

- **What are the differences between cartesian path computation and planning?**

  The cartesian path computation creates a linear trajectory between the points that are specified by the user. /It is always the same trajectory for the same points/ On the other hand, for the path planner we only provide the start and end point, and the trajectory is going to be based on the chosen planner and its priorities /optimal path or optimal time/.

- **Is there any chance that the object will be moved by robot before grasping? Why yes/no?**

  No, the object will not be moved. The gripper is opened before the trajectory from pre-grasp to grasp, so the hand cannot move the object before grasping. Additionally, the MoveIt takes the physical boundaries into account, if a trajectory cannot be executed, it won't happen.

- **Can robot collide with itself during execution of computed pick-and-place path?**

  No, the robot cannot, because the path planner takes the boundaries of the robot into account. Furthermore, for the cartesian planner, we provide the start and the end point, and the path between them are linear. Additionally, the MoveIt takes the physical boundaries into account, if a trajectory cannot be executed, it won't happen.

- **Estimated time spent on this exercise.**

  16 hours.